D 100190 **(Pages : 4)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2024

(CUCBCSS—UG)

Mathematics

MAT 6B 13 (E02)—LINEAR PROGRAMMING

(2018 Admission only)

Time : Three Hours Maximum : 80 Marks

Section A

Answer all the **twelve** *questions.*
Each question carries 1 *mark.*

1. Define convex cone of a set.
2. Show that a hyperplane in $\mathbb{R}^n$ is a convex set.
3. State graphical solution algorithm for an LPP involving two variables.
4. What is an infeasible solution ?
5. When does the simplex method indicate that the LPP has an unbounded solution.
6. Write the dual of the following LPP :

   Maximize $z = x_1 + x_2 + x_3$
   subject to the constraints $4x_1 + 3x_2 + x_3 = 6$,
   $x_1 + 2x_2 + 5x_3 = 4$,
   $x_1, x_2, x_3 \geq 0$.
7. State minimax theorem.
8. What is a transportation problem.
9. Write the following LPP in the standard form :

   Maximize $f = 5x_1 + 2x_2 - x_3$,
   subject to $2x_1 - x_2 + 6x_3 \leq 6$,
   $3x_1 + 5x_2 - 2x_3 \geq 3$
   $x_1 + 5x_2 + x_3 = 6$,
   $x_1 \geq 0, x_3 \geq 0, x_2$ unrestricted in sign.

Turn over

10. Give a brief outline of the procedure for solving a transportation problem.

11. Discuss the Hungarian method of solving an assignment problem.

12. Explain north west corner method of the transportation problem.

(12 × 1 = 12 marks)

Section B

Answer any **nine** *out of twelve questions.*
Each question carries 2 *marks.*

13. Formulate the following problem as a Linear Programming Problem : A calculator company produces a handheld calculator and a scientific calculator. Long-term projections indicate an expected demand of at least 150 scientific and 100 handheld calculators each day. Because of limitations on production capacity, no more than 250 scientific and 200 handheld calculators can be made daily. To satisfy a shipping contract, a minimum of 250 calculators must be shipped each-day. If-each scientific calculator sold results in a 20 rupees loss, but each handheld calculator produces a 50 rupees profit ; then how many of each type should be manufactured daily to maximize the net profit ?

14. Examine whether the set $S = \{(x_1, x_2) : x_1, x_2 \geq 1, x_1 \geq 0, x_2 \geq 0)\}$ is convex or not

15. Write the characteristics of standard form of LPP.

16. Explain feasible and basic feasible solutions of an LPP.

17. Verify minimax theorem for the function $f(x) = \{9, 7, 5, 3, 1\}$.

18. State the general rules for converting any primal into its dual.

19. Explain primal and dual problem.

20. Write all the steps of Vogel's approximation method for solving transportation problem.

21. How do you recognize optimality in simplex method.

22. Prove that every loop in a transportation table has an even number of cells.

23. Find an initial basic feasible solution of the following problem using north west corner rule :

	D_1	D_2	D_3	D_4	Supply
O_1	5	3	6	2	19
O_2	4	7	9	1	37
O_3	3	4	7	5	34
Demand	16	18	31	25	

24. Explain the difference between standard form and canonical form of an LPP.

(9 × 2 = 18 marks)

Section C

Answer any **six** *out of nine questions.*
Each question carries 5 *marks.*

25. Show that the set of all convex combinations of a finite numbers of points $x_1, x_2, \ldots, x_k$ is a convex set.

26. Solve graphically the following LPP :

Maximize $5x_1 + 3x_2$
subject to $x_1 + x_2 \leq 6$,
$2x_1 + 3x_2 \geq 6$,
$0 \leq x_1 \leq 4$,
$0 \leq x_2 \leq 4$.

27. How do you check from the transportation table that a feasible solution is basic or not ?

28. Solve the following LPP by simplex method :

Maximize $f(x) = x_1 + 2x_2$
subject to : $x_1 + 2x_2 \leq 3$,
$x_1 + x_2 \leq 2$,
$x_1 \leq 1$,
$x_1 \geq 0$,
$x_2 \geq 0$.

29. Explain the Charne's Big -M method.

30. Prove that the dual of the dual is primal.

31. What is feasible solution and non degenerate solution in transportation problem ?

32. Explain Hungarian method.

Turn over

33. Explain Vogel's approximation method by obtaining initial feasible solution of the following transportation problem.

	D_1	D_2	D_3	D_4	Availability
O_1	5	8	3	6	30
O_2	4	5	7	4	50
O_3	6	2	4	6	20
Requirement	30	40	20	10	

(6 × 5 = 30 marks)

Section D

Answer any **two** *out of three questions.*
Each question carries 10 *marks.*

34. Let $A \subseteq \mathbb{R}^n$ be any set. Prove that, the convex hull of A is the set of all finite convex combination of vectors in A.

35. Find the optimal solution for the assignment problem with the following cost matrix :

		Area			
		1	2	3	4
Salesman	P	11	17	8	16
	Q	9	7	12	6
	R	13	16	15	12
	S	14	10	12	11

36. Solve the following problem by simplex method. Also solve it by solving its dual graphically.

$z = 5x_1 - 3x_2 + 4x_3$
subject to $x_1 - x_2 \leq 1$,
$-3x_1 + 2x_2 + 2x_3 \leq 1$,
$4x_1 - x_3 = 1$,
$x_2 \geq 0$,
$x_3 \geq 0$,
x_1, unrestricted in sign.

(2 × 10 = 20 marks)